Half-degree reciprocal constraints with exact phases

T-1862 · 2026-11-07 · Prime-field first sign-mode constraint

When 4m divides r−1, write h=2mn. Binomial symmetry makes F_0 reciprocal of degree2n. The substitution $W=v+v^{-1}$ gives $F_0(v)=v^nH(W)$, where H has degree n. Balanced affine Clenshaw composition builds H and D_n from the upper n+1 streamed coefficients, without expanding F_0.

Eligibility: 4m divides r−1; F_0 has degree 2n

Upper n+1 coefficients ⟶ balanced Clenshaw ⟶ H(W), $D_n(W)$
$F_0(v) = v^n H(v + v^{-1})$ · both output polynomials have degree n

Four phases u: $\gcd(D_n - u - u^{-1}, uH - c)$

Capped quadratic lift ⟶ gcd with $v^n - u$ ⟶ disjoint union + zero

Scale back to Z ⟶ exact first-constraint roots ⟶ existing recovery

Unsupported or capped: original path, with all attempted work charged

Exact phase transfer. On the nonzero image domain $v^{(4n)}=1$. For each u in {1,−1,i,−i}, form $\gcd(D_n(W)-u-u^{-1}, uH(W)-c)$. Lift surviving factors through v^2-Wv+1 and intersect with v^n-u using modular powering. This preserves orientation and removes ramified multiplicity. Add zero exactly when $f_0=A$, then substitute v=tZ and normalize. The result equals the original first-constraint gcd.

Native exact check	Count
Configurations / nonzero identity values	27 / 4,496
Branch / lifted point checks	308,844 / 305,632
Exact full image polynomials	2,409
Wrong orientations rejected / ramified incidences	328 / 31
Lift-cap rejection controls	183

Integrated public recovery. All281targets on F257, $y^2=x^3+x+7$, base(1,3), order281, mode5 match isolated scalar/routes/ statistics:2,881source calls, zero unresolved. At the existing seven-element lift cap,10of275compressed attempts complete and265fall back. Fallback costs are additional. The inherited617-target panel remains5,929calls, zero unresolved.

Costs and limits. Balanced composition has O(M(n) log n) polynomial work plus coefficient generation; four degree-n gcds remain. H and D_n retain2(n+1)live values, with capacities/metadata charged. Temporary bound O(n log n+B); capped lifts, modular powers, inversions and up to64public quartic-root search exponentiations are charged. Half degree is not a total-work or timing claim. Unsupported parameters and exceeded lift/union caps use the original constraint.

Evidence. validation01 pins54transitive inputs; native overflow checks pass. Unsigned producer evidence. No cloud/timing trial or fullCargo rerun; original4,563-case cohort unchanged. DERIVATION.md provides proof, replay, controls and historical limitations.

Source and next construction. Bhargava–Zieve, Factoring Dickson polynomials over finite fields, websites.umich.edu/ zieve/ papers/manjul.pdf; exact predecessor code. KB unavailable; no novelty claim. Next retain compressed phase-tagged states through later modes, avoiding the early lift that triggered265fallbacks.